Bilkent University

Department of Electrical and Electronics Engineering

EEE-313 Project Design Phase Report

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[ID]

"I affirm that I have not given or received any unauthorized help on this report and that this work is my own."

Introduction:

The aim of this project is to create a single-supply thermocouple amplifier that will generate a voltage difference proportional to the temperature at the output. Using a K-type thermocouple, OPAMPS, a BJT and a LED to show the situation. The thermocouple will create small voltage differences with the temperature. To measure the difference in a better way, we will amplify this voltage using OPAMP amplifiers. We will then adjust the heater by switching it ON and OFF. This way, the temperature of the resistor will be kept stable. The resistor will be kept at 36℃ since $T_s = 30 + mod([ID], 40)$℃ where [ID] is my Bilkent ID. When the heater is ON, the LED must turn ON for visual feedback.

Design Specifications:

Firstly, I found the corresponding voltage value for my T_s value. For every 1°C increase, the the voltage changes by 39.2μV. So the value can be calculated by

$$\frac{39.2\mu V}{°C} \times 36°C = 1411.2\mu V = 1.4112mV$$

When it reaches to this voltage, the heater and therefore the LED must be turned OFF.

Here is my overall circuit design in LTSpice:

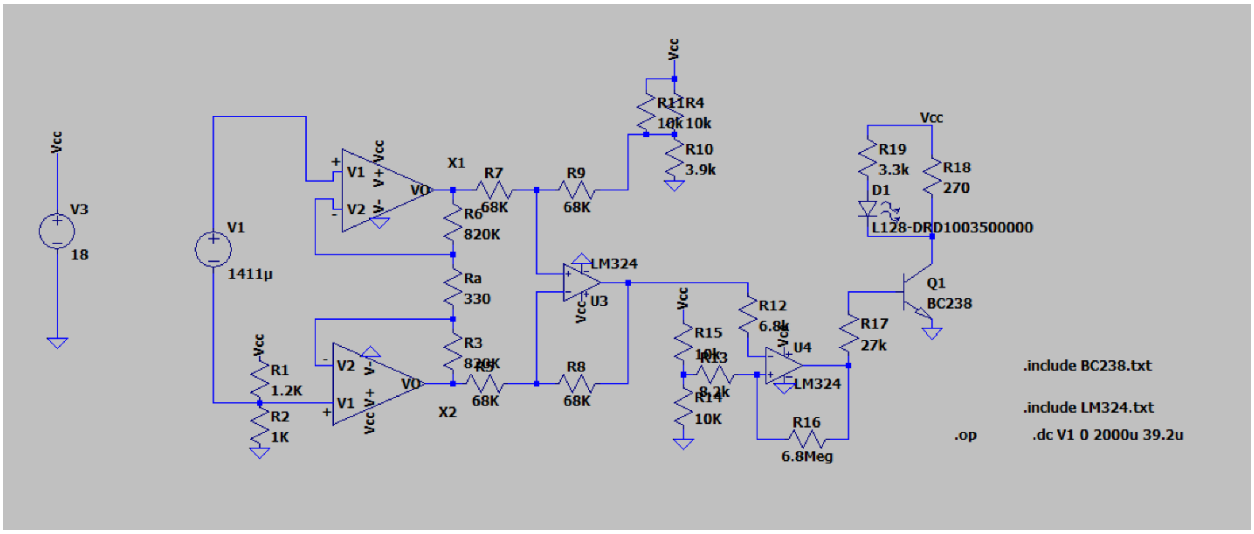


Figure 1: Overall Circuit Design in LTSpice

To briefly explain the circuit, we have a Instrumentation Amplifier to amplify the input voltage. This is a high-gain amplification therefore we must take the offset values of the LM324 OPAMPS into account. How the offset values are calculated will be explained in the later parts of this report. Then, we get this output to a comparator OPAMP. This will compare the coming output to the desired voltage value which is created by a voltage divider circuit using Vcc. Then, the output of the comparator circuit will be connected to a BJT to use it as a switch. It will be changing states SAT and OFF to determine the status of the heater and the LED.

Measuring the offset voltages:

The offset values of the first OPAMP.

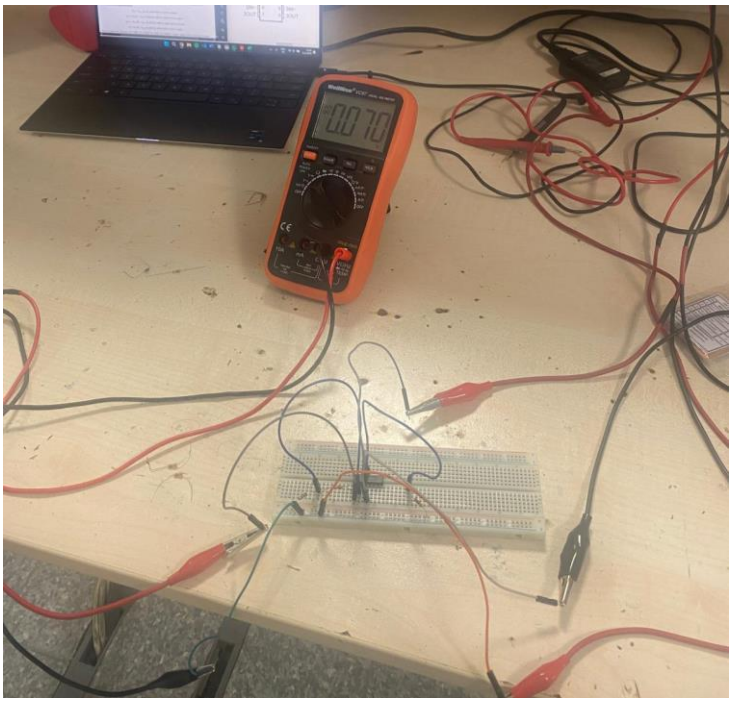


Figure 2:Vout1

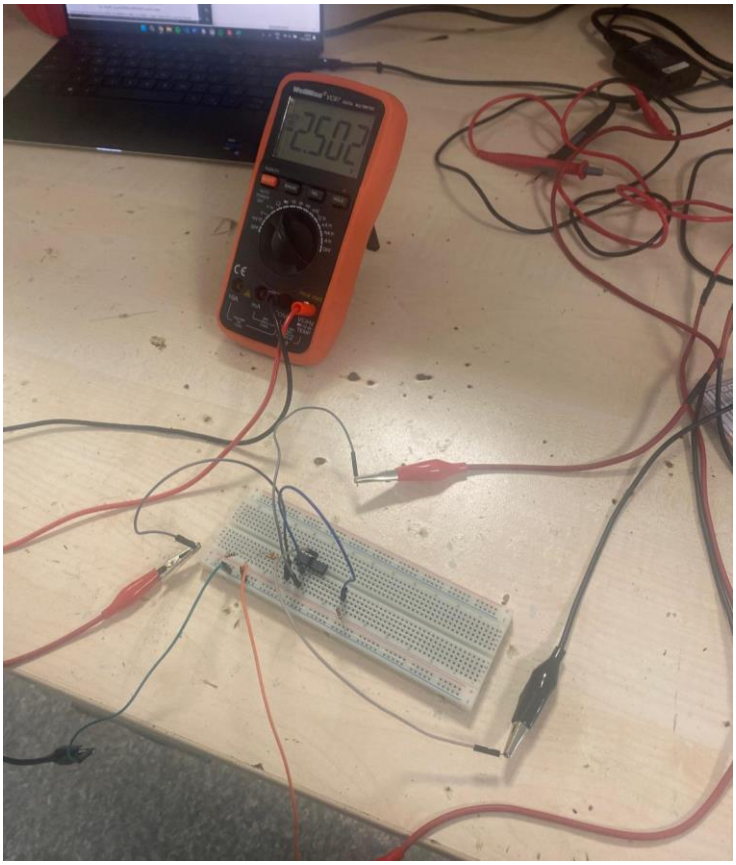


Figure 3:Vout2

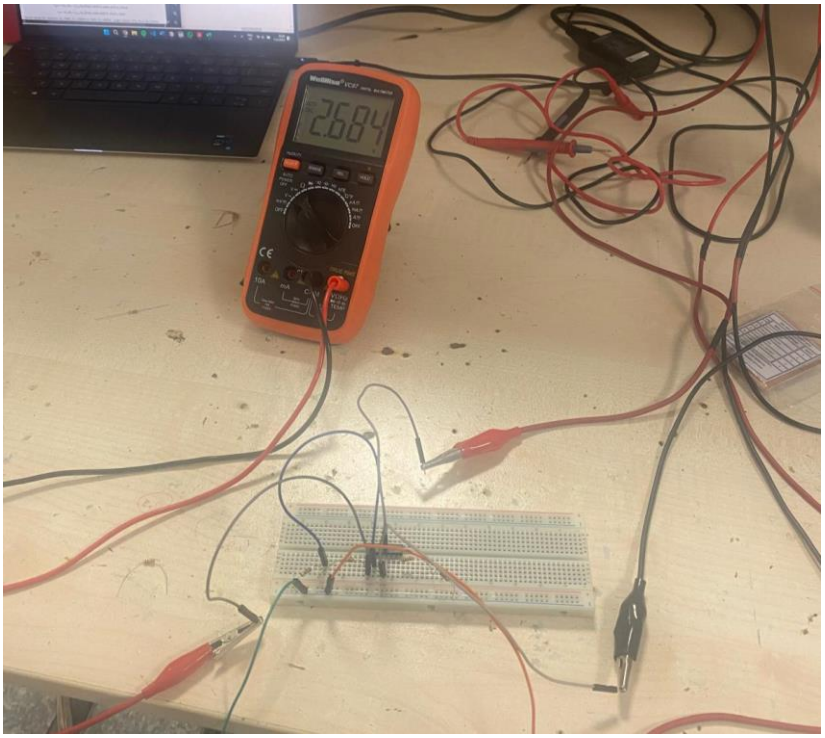


Figure 4:Vout3

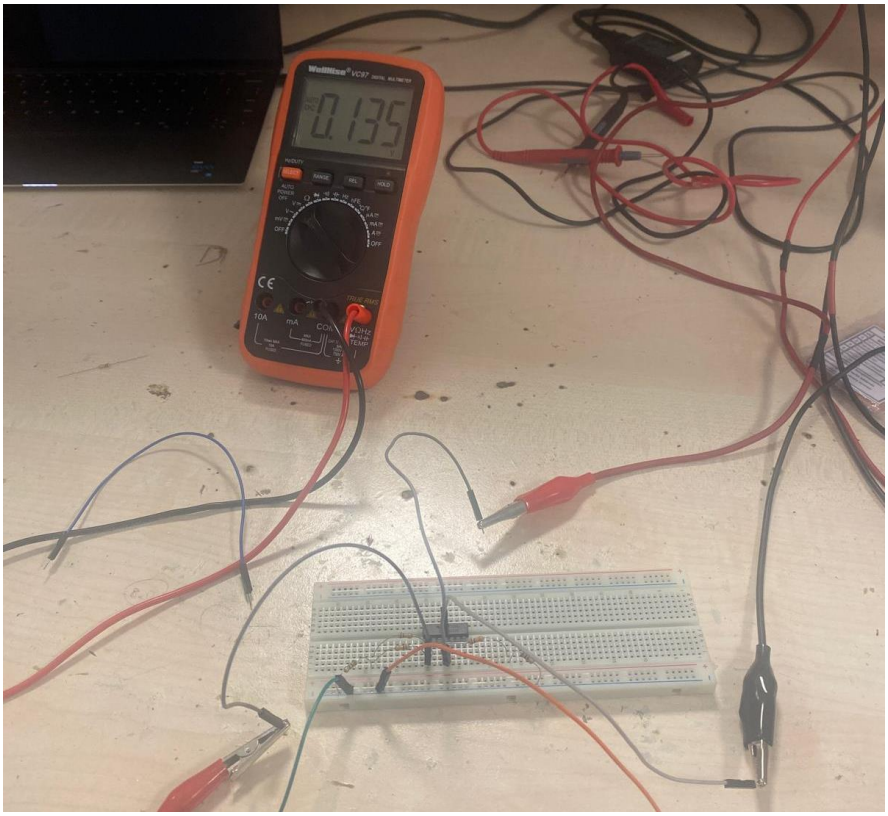


Figure 5:Vout4

R1	100			
R2	1,00E+05			
R3	1,00E+05			
			S1	S2
vout1	0,07		ON	ON
vout2	-2,502		ON	OFF
vout3	2,684		OFF	ON
vout4	0,135		OFF	OFF
VIO	-7,00E-05	-0,07	mV	
IB2	-2,57E-08	-25,72	nA	
IB1	-2,61E-08	-26,14	nA	
IIO	-6,50E-10	-0,65	nA	
IB1-IB2	-4,20E-10	-0,42	nA	

Figure 6: Offset Values Found From Excel

The offset values of the second OPAMP:

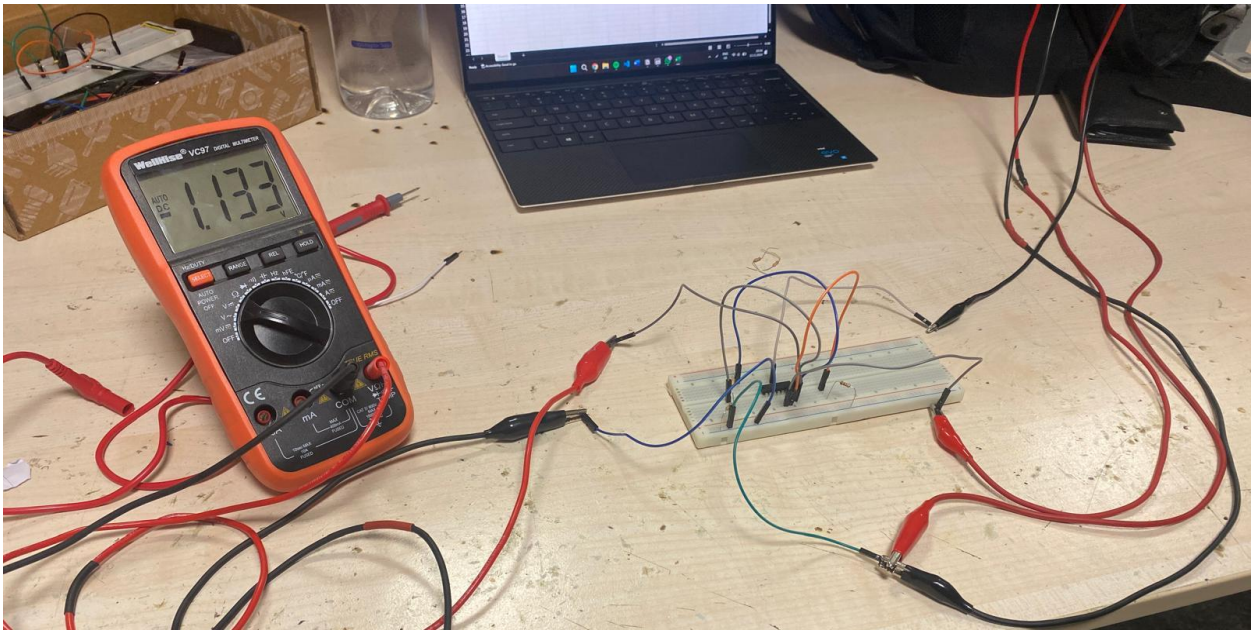


Figure 7:Vout1

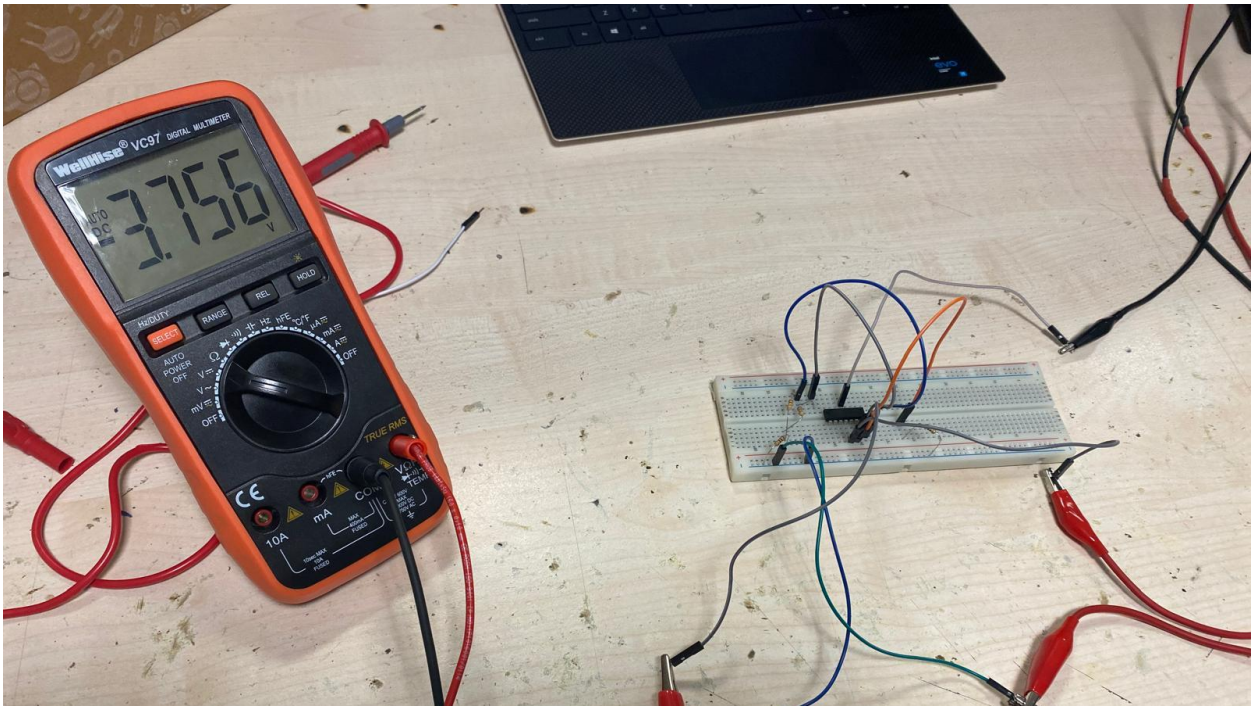


Figure 8:Vout2

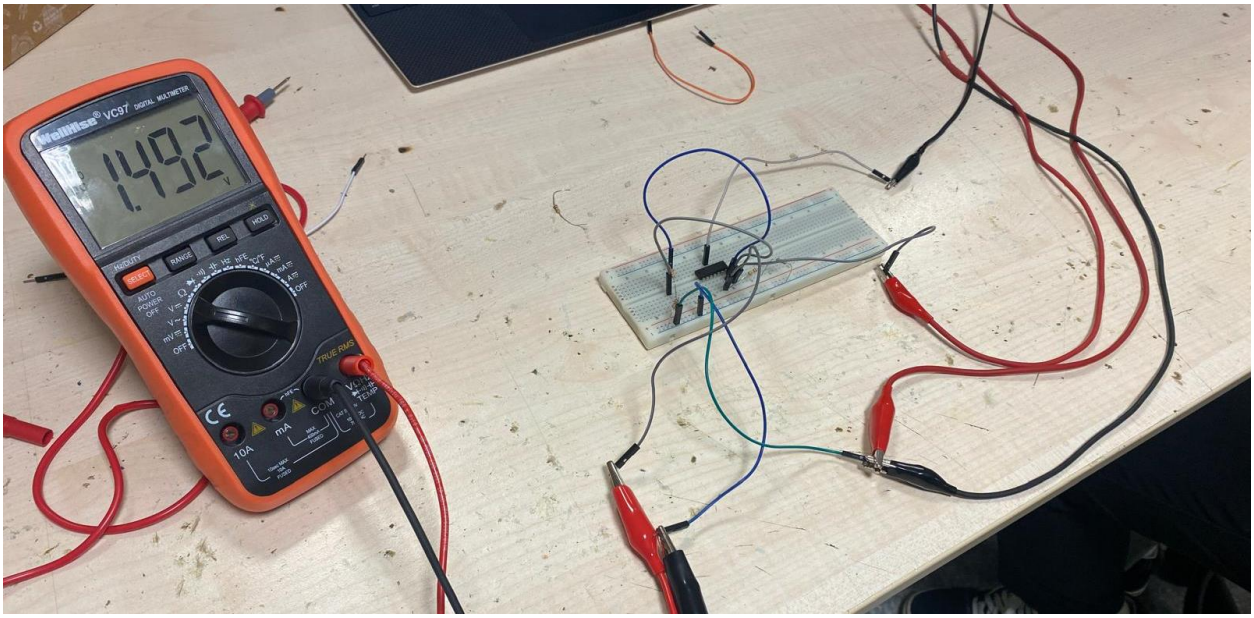


Figure 9:Vout3

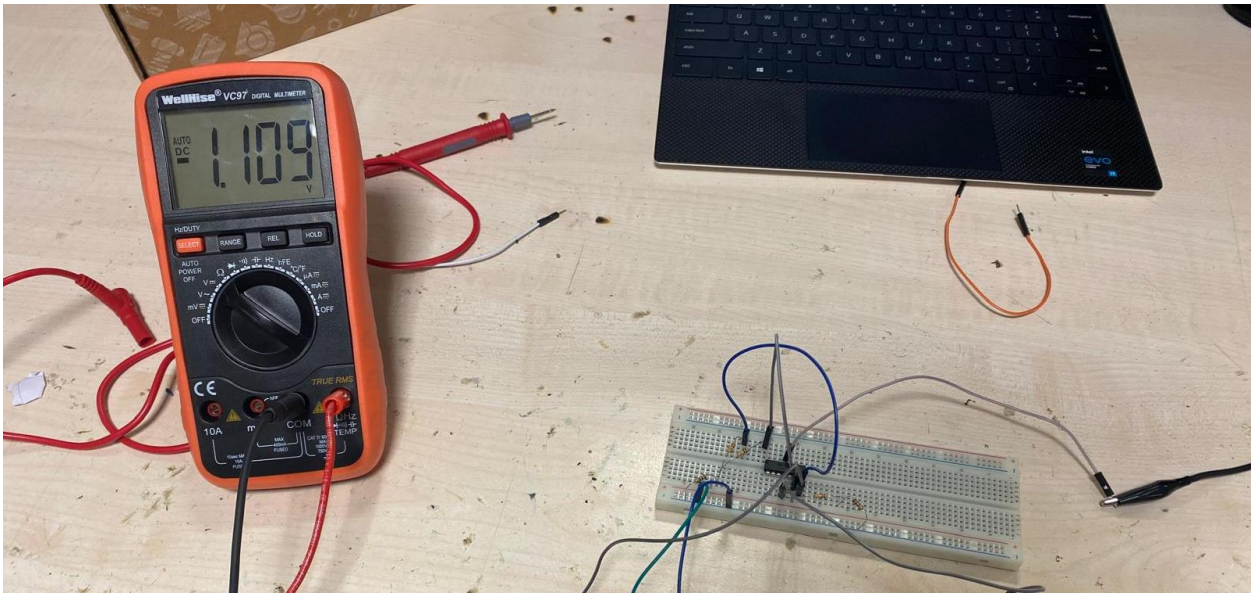


Figure 10:Vout4

R1	100			
R2	1,00E+05			
R3	1,00E+05			
			S1	S2
vout1	-1,133		ON	ON
vout2	-3,756		ON	OFF
vout3	1,492		OFF	ON
vout4	-1,109		OFF	OFF
VIO	1,13E-03	1,13	mV	
IB2	-2,62E-08	-26,23	nA	
IB1	-2,63E-08	-26,25	nA	
IIO	-2,40E-10	-0,24	nA	
IB1-IB2	-2,00E-11	-0,02	nA	

Figure 11:Offset values found from Excel

To use these values, I used the OPAMP circuit given to us on Moodle. I used the OPAMP with the offset values on my project. Here is one of the images of the circuit.

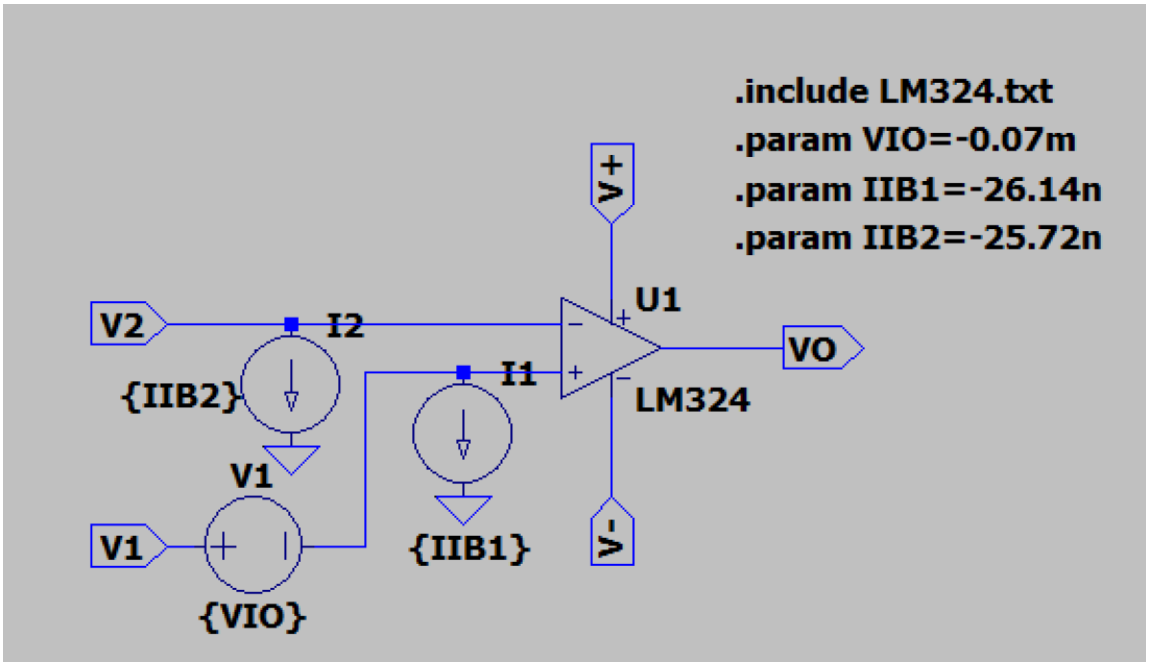


Figure 12: Schematic of Offset OPAMP circuit on LTSpice

Desired Specifications and Results:

1) **The output voltage should be 2V∓0.5V when the input voltage is 0V (the termocouple at room temperature).**

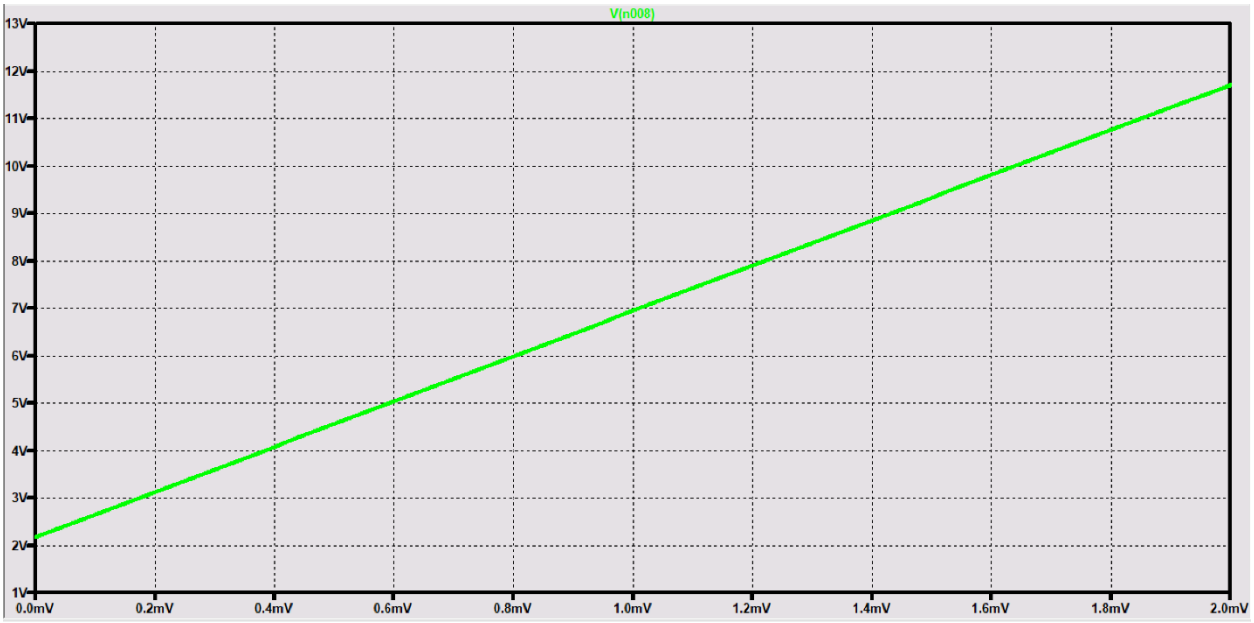


Figure 13: Plot of the output voltage with increasing Vthermocouple

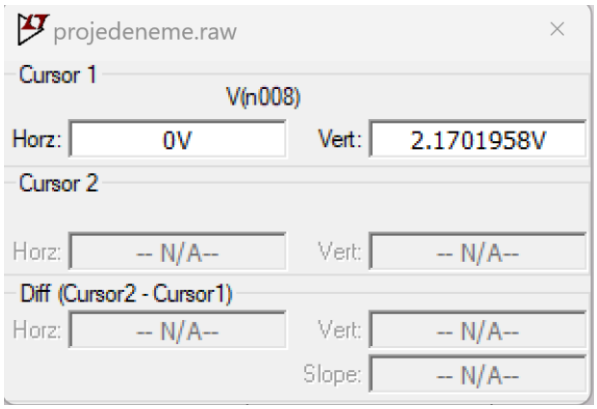


Figure 14: Voltage Value When V=0

As it can be seen from the plots, Vout is equal to 2.1701V which is inside our boundaries. This satisfies the first specification.

2) **The output voltage should be 9V∓1V when the input voltage is at $39.2\mu Vx36 = 1.4112mV$ (temperature is at T_S).**

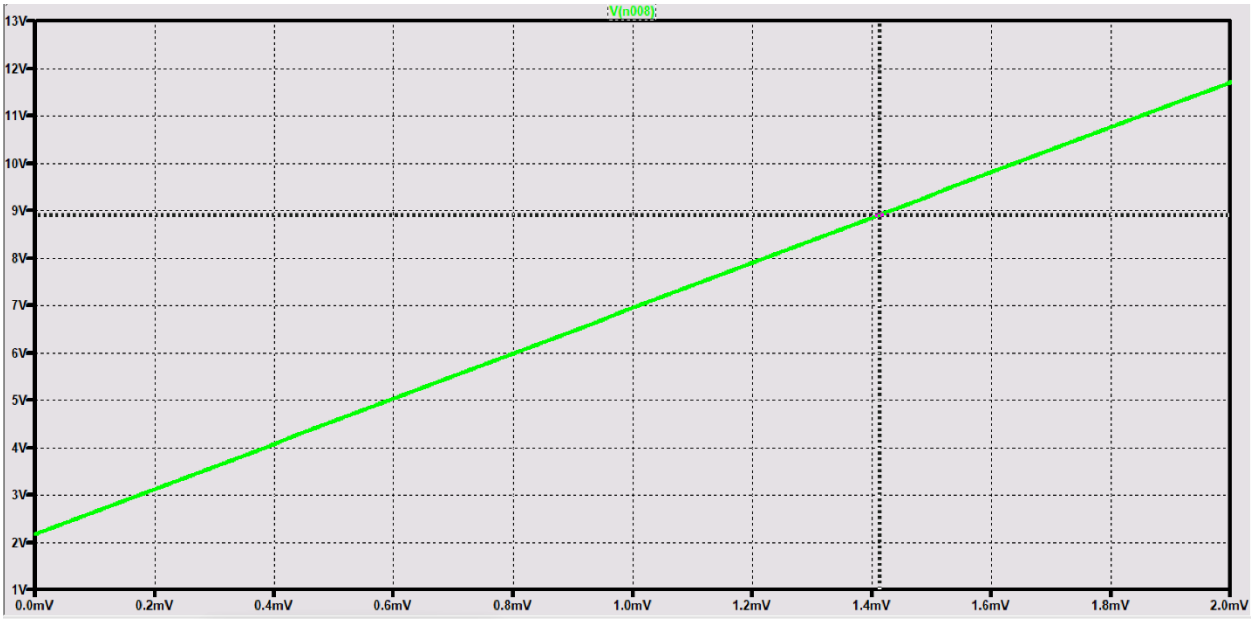


Figure 15:Plot of the output voltage with increasing Vthermocouple

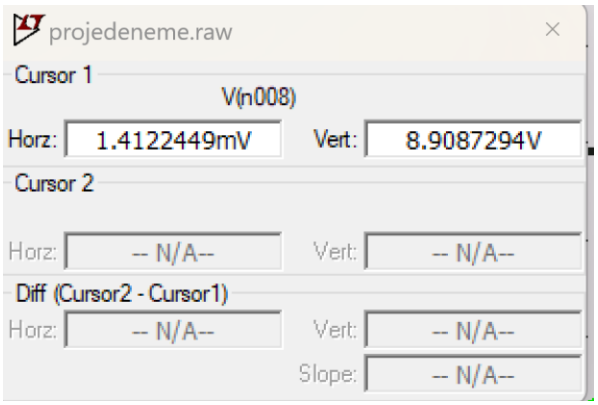


Figure 16: Output Voltage When V=1.4112mV

As it can be seen from the plots, Vout is equal to 8.9087V which is in our boundaries. So this satisfies this specification.

3) **LED should turn OFF when voltage passes 9V and it should be ON otherwise.**

To show this, we can look at the current passing on the LED. To see it in a better way, I made the plot with increasing Vthermocouple to see where the LED goes OFF.

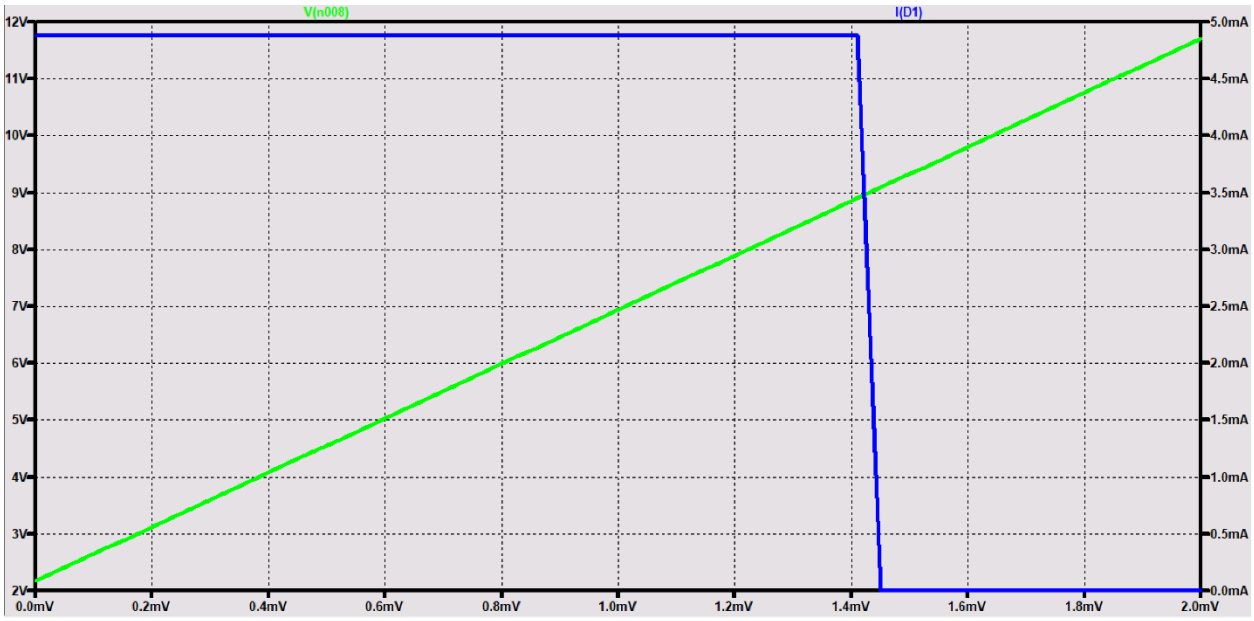


Figure 17: Plot of the current on the LED with increasing V

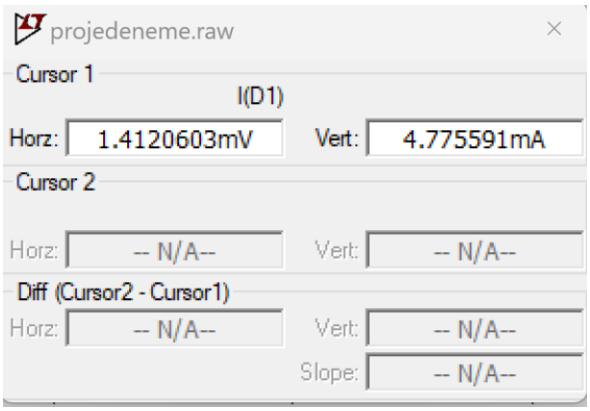


Figure 18: Current when V=1.412mV

As it can be seen on the plots, when Vout passes 9V, where Vthermocouple is 1.4121mV, the current on the LED starts to drop to 0. Therefore it is OFF. Otherwise, there is a flow of current on the LED which is around 4.77mA which states the LED is ON. Therefore, this specification is also satisfied.

Conclusion:

To conclude, I successfully simulated and designed the temperature controlled heating system project. All of the 3 specifications are satisfied as shown. A better vision and understanding for OPAMP circuits have been gained. The errors are caused by the offsets therefore the offset values are important and adjusted in the circuit. This simulation now shows that our circuit is ready for physical implementation and we can start to work on our PCB designs.